19

Temperature

EQUATIONS AND CONCEPTS

The **Celsius temperature** T_C is related to the absolute temperature T (in kelvins, K) according to Equation 19.1. *The size of a degree (unit change in temperature) on the Celsius scale equals the size of a degree on the kelvin scale.*

$$T_C = T - 273.15 \quad (19.1)$$

$$0°\text{C} = 273.15 \text{ K}$$

Conversion between Fahrenheit and Celsius temperature scales is based on the freezing point temperature and boiling point temperature of water set for each scale.

$$T_F = \tfrac{9}{5}\, T_C + 32°\text{F} \quad (19.2)$$

$$T_C = \tfrac{5}{9}\left(T_F - 32°\text{F}\right)$$

Freezing point = 0°C = 32°F

Boiling point = 100°C = 212°F

Thermal expansion (or contraction) due to an increase in temperature, results in changes in length, surface area, and volume of solids and a change in the volume of liquids. The fractional change is proportional to the change in temperature and to a parameter (coefficient of expansion) which is characteristic of a particular material.

Length expansion (solids)

$$\Delta L = \alpha L_i \Delta T \quad (19.4)$$

$$L = L_i\,(1 + \alpha \Delta T)$$

Area expansion (solids)

$$\Delta A = \gamma A_i \Delta T$$

$$A = A_i(1 + \gamma \Delta T)$$

α = coefficient of linear expansion

$\gamma \approx 2\alpha$ = coefficient of area expansion

$\beta \approx 3\alpha$ = coefficient of linear expansion

Volume expansion (solids and liquids)

$$\Delta V = \beta V_i \Delta T \quad (19.6)$$

$$V = V_i(1 + \beta \Delta T)$$

The **number of moles** in a sample of any substance (element or compound) equals the ratio of the mass of the sample to the molar mass characteristic of that particular substance. *One mole of any substance is that mass which contains Avogadro's number* (N_A) *of molecules.*

$$n = \frac{m}{M} \quad (19.7)$$

$$N_A = 6.02 \times 10^{23} \text{ particles/mole}$$

The equation of state of an ideal gas is shown in two forms: expressed in terms of the number of moles (n) and the universal gas constant (R) as stated in Equation 19.8 or expressed in terms of the number of molecules (N) and Boltzmann's constant (k_B) as stated in Equation 19.10.

$$PV = nRT \tag{19.8}$$

n = number of moles

$$R = 8.314 \text{ J/mol} \cdot \text{K} \tag{19.9}$$

$$R = 0.082\,06 \text{ L} \cdot \text{atm/mol} \cdot \text{K}$$

$$PV = Nk_BT \tag{19.10}$$

N = number of molecules

$$k_B = 1.38 \times 10^{-23} \text{ J/K} \tag{19.11}$$

REVIEW CHECKLIST

You should be able to:

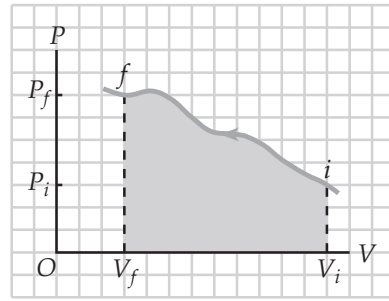
- Describe the operation of the constant-volume gas thermometer and the way in which it was used to define the absolute gas temperature scale. (Section 19.3)
- Convert among the various temperature scales: between Celsius and Kelvin, between Fahrenheit and Kelvin, and between Celsius and Fahrenheit. (Section 19.3)
- Use the thermal expansion coefficients for length, area, and volume to calculate changes in these quantities associated with changes in temperature. (Section 19.4)
- Make calculations using the ideal gas equation, expressed either in terms of n and R or in terms of N and k_B. (Section 19.5)

ANSWER TO AN OBJECTIVE QUESTION

10. A rubber balloon is filled with 1 L of air at 1 atm and 300 K and is then put into a cryogenic refrigerator at 100 K. The rubber remains flexible as it cools. **(i)** What happens to the volume of the balloon? (a) It decreases to $\frac{1}{3}$ L. (b) It decreases to $1/\sqrt{3}$ L. (c) It is constant. (d) It increases to $\sqrt{3}$ L. (e) It increases to 3 L. **(ii)** What happens to the pressure of the air in the balloon? (a) It decreases to $\frac{1}{3}$ atm. (b) It decreases to $1/\sqrt{3}$ atm. (c) It is constant. (d) It increases to $\sqrt{3}$ atm. (e) It increases to 3 atm.

Answer (i) (a) and **(ii)** (c). The tiny quantities of carbon dioxide and water in the air solidify on the way down to 100 K, but the oxygen, nitrogen, and argon remain gases, and it is reasonable to assume that their densities are low enough to permit use of the ideal gas equation of state. To qualify as a balloon, the container cannot have a fast leak, so in $PV = nRT$ we take the quantity of gas n, as well as the universal constant R, as constant in the contraction process. A refrigerator is not airtight, so the pressure inside the box stays equal to the constant atmospheric pressure outside. The flexible wall of the balloon moves to

The work done on a gas when the volume changes from V_i to V_f depends on the path taken between the initial and final states. In order to evaluate the integral in Equation 20.9, the manner in which pressure varies with volume must be known.



Work on gas = –(Area under curve)

$$W = -\int_{V_i}^{V_f} P dV \qquad (20.9)$$

The first law of thermodynamics is stated in mathematical form by Equation 20.10. The change in the internal energy of a system equals the sum of the energy transferred across the boundary by heat and the energy transferred by work. *The values of both Q and W depend on the path taken between the initial and final states;* ΔE_{int} *is path independent.*

$$\Delta E_{int} = Q + W \qquad (20.10)$$

Q = the energy transferred to the system by heat.

W = the work done on the system.

ΔE_{int} = change in internal energy.

Applications of the laws of thermodynamics will be described using some of the terms and equations described and shown below.

An **isolated system** does not interact with the surroundings. *The internal energy of an isolated system remains constant.*

Isolated system

$Q = W = 0$

$\Delta E_{int} = 0$

In a **cyclic process** a system is returned to the initial state and there is no change in the internal energy of the system.

Cyclic process

$Q = -W$

$\Delta E_{int} = 0$

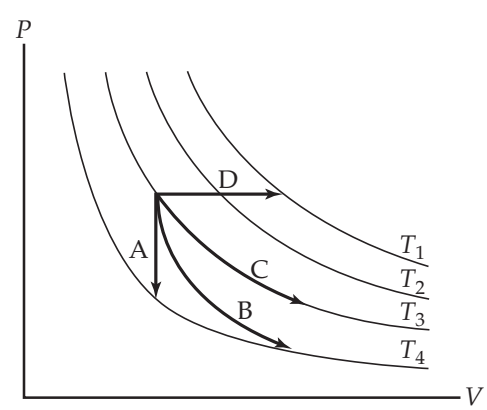
***PV* diagrams for basic thermal processes** are illustrated in the figure at right.

A = isovolumetric process

B = adiabatic process

C = isothermal process

D = isobaric process



Basic thermal processes

A system can undergo an **adiabatic process** if it is thermally insulated from its surroundings.

Adiabatic process

$$E_{\text{int}} = W \tag{20.11}$$

In an **isobaric process** energy is transferred into (out of) the gas as it expands (contracts).

Isobaric process

$$W = -P(V_f - V_i) \tag{20.12}$$

W = work done on the gas

In an **isovolumetric process** zero work is done and the net energy added to the system by heat at constant volume increases the internal energy of the system.

Isovolumetric process

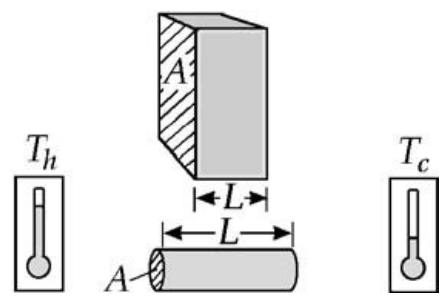
$$E_{int} = Q \tag{20.13}$$

In an **isothermal process** no change occurs in the internal energy of a system; any energy that enters the system by heat is transferred out of the system by work, ($Q = -W$).

Isothermal process

$$W_{\text{env}} = nRT \ln\left(\frac{V_f}{V_i}\right) \tag{20.14}$$

Rate of energy transfer by conduction through a slab or rod of material is proportional to the temperature difference between the hot and cold faces and inversely proportional to the thickness of the slab or length of the rod (see figure). The thermal conductivity, k, is characteristic of a particular material; *good thermal conductors have large values of k.* The units of k are J/s · m · °C



$$P = kA\left(\frac{T_h - T_c}{L}\right) \tag{20.16}$$

Rate of energy conduction through a compound slab of area A is found by using Equation. 20.17. The temperatures of the outer surfaces are T_c and T_h. The thicknesses of the slabs are $L_1, L_2, L_3, \ldots$ and $k_1, k_2, k_3, \ldots$ are the respective thermal conductivities.

$$P = \frac{A(T_h - T_c)}{\sum_i (L_i/k_i)} \tag{20.17}$$

The ***R*-value** of a material is the ratio of thickness to thermal conductivity. *Materials with large R-values are good insulators.* In the US, R-values are given in engineering units: ft² · °F · h/BTU.

$$R_i = \frac{L_i}{k_i}$$

Using the definition of R as stated above, Equation 20.17 can be written as shown in Equation 20.18.

$$P = \frac{A(T_h - T_c)}{\sum_i R_i} \tag{20.18}$$

The **rate of energy transfer by radiation** (radiated power) from a surface of area A and temperature T is given by Stefan's law, Equation 20.19. The surface temperature T is measured in kelvins and the value of e (the emissivity of the radiating body) is determined by the nature of the surface.

$$P = \sigma A e T^4 \tag{20.19}$$

$$\sigma = 5.669\,6 \times 10^{-8}\ \mathrm{W/m^2 \cdot K^4}$$

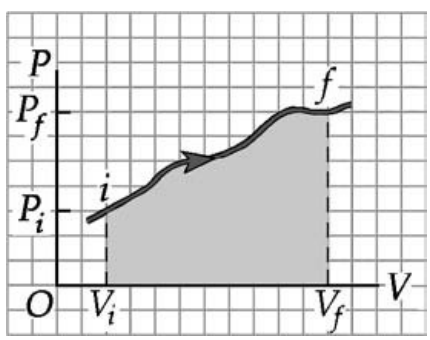
e = emissivity (values from 0 to 1)

Net radiated power by an object at temperature T, in an environment at temperature T_0, is the difference between the rate of emission to the environment and rate of absorption. *At equilibrium $T = T_0$, the two rates are equal, and the temperature of the object remains constant.*

$$P_{\text{net}} = \sigma A e \left(T^4 - T_0^4\right) \tag{20.20}$$

SUGGESTIONS, SKILLS, AND STRATEGIES

Many applications of the first law of thermodynamics deal with the work done on a system that undergoes a change in state. Consider the PV diagram showing the path taken by a gas from an initial state with pressure and volume values of P_i and V_i to a final state with pressure and volume values of P_f and V_f.



If the pressure is a known function of the volume, the work done on the gas can be calculated directly using Equation 20.9. If the path taken by the gas from its initial to its final state can be shown in a PV diagram, the work done on the gas during the expansion is equal to the negative of the area under the PV curve (the shaded region) shown in the figure. *It is important to recognize that the work depends on the path taken as the gas goes from i to f.*

PROBLEM SOLVING STRATEGY: CALORIMETRY PROBLEMS

When solving calorimetry problems, consider the following points:

- Be sure your units are consistent throughout. For instance, if you are using specific heats in cal/g · °C, be sure that masses are in grams and temperatures are Celsius throughout.
- Energy losses and gains are found by using $Q = mc\Delta T$ only for those intervals during which no phase changes occur. The equations $Q = \Delta mL_f$ and $Q = \Delta mL_v$ are to be used only when phase changes are taking place.

21

The Kinetic Theory of Gases

EQUATIONS AND CONCEPTS

The **pressure of an ideal gas** is proportional to the number of molecules per unit volume and the average translational kinetic energy per molecule.

$$P = \frac{2}{3}\left(\frac{N}{V}\right)\left(\frac{1}{2}m_o\overline{v^2}\right) \quad (21.2)$$

m_0 = molecular mass

The average kinetic energy per molecule in a gas is proportional to the absolute temperature. *This result follows from Equation 21.2 and the equation of state for an ideal gas, $PV = Nk_BT$.*

$$\frac{1}{2}m_o\overline{v^2} = \frac{3}{2}k_BT \quad (21.4)$$

$\overline{v^2}$ = root-mean-square speed

The **internal energy** of a sample of an ideal monatomic gas depends only on the absolute temperature. *In Equation 21.10, N is the number of molecules and n is the number of moles.*

$$E_{\text{int}} = K_{\text{tot trans}} = \frac{3}{2}Nk_BT = \frac{3}{2}nRT \quad (21.10)$$

(monatomic gas)

The **molecular specific heat of an ideal gas** is defined for two processes:

C_V (specific heat at constant volume)
C_P (specific heat at constant pressure)

At constant volume

$$Q = nC_V\Delta T \quad (21.8)$$

$$C_V = \frac{3}{2}R \quad \text{(monatomic)} \quad (21.14)$$

$$C_V = \frac{5}{2}R \quad \text{(diatomic)} \quad (21.21)$$

At constant pressure

$$Q = nC_P\Delta T \quad (21.9)$$

$$C_P = \frac{5}{2}R \quad \text{(monatomic)}$$

The energy transferred by heat required to take a gas between two given temperatures at constant pressure will be greater than that required to take the gas between the same two temperatures at constant volume; $C_P > C_V$. See *Suggestions, Skills, and Strategies* for an explanation of this result.

An adiabatic processes for an ideal gas involves changes in the variables P, V, and T. During the adiabatic process, no energy is transferred by heat between the gas and its environment.

$$PV^{\gamma} = \text{constant} \tag{21.18}$$

$$TV^{\gamma-1} = \text{constant} \tag{21.20}$$

The **Boltzmann distribution law** (which is valid for a system containing a large number of particles) states that the probability of finding the particles in a particular arrangement (distribution of energy values) varies exponentially as the negative of the energy divided by k_BT. *In Equation 21.23, $n_V(E)$ is a function called the number density.*

$$n_V(E) = n_0 e^{-E/k_BT} \tag{21.23}$$

$n_V(E)\,dE$ = number of molecules per unit volume with energy within the range between E and $E + dE$.

The **Maxwell–Boltzmann speed distribution function** describes the most probable distribution of speeds of N gas molecules at temperature T (where k_B is the Boltzmann constant). *The probability that a molecule has a speed in the range between v and $v + dv$ is equal to dN/N.*

$$N_v = 4\pi N\left(\frac{m_o}{2\pi k_BT}\right)^{3/2} v^2 e^{-m_o v^2/2k_BT} \tag{21.24}$$

$dN = N_v\,dv$

dN = number of molecules with speeds between v and $v + dv$.

Root-mean-square speed, **average speed**, and **most probable speed** can be calculated from the Maxwell–Boltzmann distribution function. At a given temperature, lighter molecules have higher average speeds than heavier ones.

$$v_{\text{rms}} = 1.73\sqrt{\frac{k_BT}{m_o}} \tag{21.25}$$

$$v_{\text{avg}} = 1.60\sqrt{\frac{k_BT}{m_o}} \tag{21.26}$$

$$v_{\text{mp}} = 1.41\sqrt{\frac{k_BT}{m_o}} \tag{21.27}$$

SUGGESTIONS, SKILLS, AND STRATEGIES

CALCULATING SPECIFIC HEATS

It is important to recognize that, depending on the type of molecule (and hence the number of degrees of freedom), the ratio of specific heats for a gas (γ) might be 1.67, 1.40, or some

other value. However, in any ideal gas, two equations are always valid, and can be used to determine the specific heats C_P and C_V from a known value of γ.

$$\gamma = \frac{C_P}{C_V} \quad (21.17) \qquad \text{and} \qquad C_P - C_V = R \quad (21.16)$$

Combining the equations above, we find

$$C_V = \frac{R}{\gamma - 1} \qquad \text{and} \qquad C_P = \frac{\gamma R}{\gamma - 1}$$

These equations may be used for all ideal gases.

C_P is greater than C_V for the following reason. The temperature of a sample of gas can be increased by a given amount ΔT by either a constant pressure process or by a constant volume process. The resulting increase in internal energy of the gas, ΔE_{int} will be the same in both cases. When energy is transferred to a gas by heat at constant pressure, the volume increases and therefore a portion of the added energy leaves the system by work. Only the remaining portion of the energy transferred by heat contributes to an increase in the internal energy of the gas. If the temperature increase is accomplished by a constant volume process, no work is done; and all of the energy transferred by heat contributes to an increase in the internal energy of the gas. Therefore the added energy per mole required to effect the same increase in internal energy, and hence the same temperature increase, must be greater for the constant pressure process.

REVIEW CHECKLIST

You should be able to:

- State and understand the assumptions made in developing the molecular model of an ideal gas. (Section 21.1)
- Calculate pressure, average translational kinetic energy per molecule and rms speed per molecule and quantity (moles) of a confined gas. (Section 21.1)
- Calculate Q and ΔE_{int} when the temperature of a given number of moles of a gas is increased by an amount ΔT through a constant volume or a constant pressure process. (Section 21.2)
- Make calculations involving an adiabatic process for an ideal gas. (Section 21.3)
- Find values for average speed, root-mean-square speed, and most probable speed when given the individual values of speed for a number of particles. (Section 21.5)

22

Heat Engines, Entropy, and the Second Law of Thermodynamics

EQUATIONS AND CONCEPTS

The **net work done by a heat engine** during one cycle equals the net energy, Q_{net}, transferred to the engine. In Equation 22.1 $|Q_h|$ is the quantity of energy absorbed from a hot reservoir and $|Q_c|$ is the amount of energy expelled to the cold reservoir.

$$W_{eng} = |Q_h| - |Q_c| \tag{22.1}$$

The **thermal efficiency**, e, of a heat engine is defined as the ratio of the net work done by the engine to the energy input at the higher temperature during one cycle. *For any real engine,* $e < 1$.

$$e \equiv \frac{W_{eng}}{|Q_h|} = \frac{|Q_h| - |Q_c|}{|Q_h|} = 1 - \frac{|Q_c|}{|Q_h|} \tag{22.2}$$

The **effectiveness of a heat pump** is described in terms of the coefficient of performance, COP. W = the work done on the heat pump.

$$\text{COP (cooling mode)} = \frac{|Q_c|}{W} \tag{22.3}$$

$|Q_c|$ = energy transferred at low temperature

$$\text{COP (heating mode)} = \frac{|Q_h|}{W} \tag{22.4}$$

$|Q_h|$ = energy transferred at high temperature

The **Carnot efficiency** is the maximum theoretical efficiency of a heat engine operating reversibly in a cycle between T_c and T_h. *No real heat engine operating irreversibly between two reservoirs* $(T_c$ *and* $T_h)$ *can be more efficient than a Carnot engine operating between the same two reservoirs.*

$$e_C = 1 - \frac{T_c}{T_h} \tag{22.6}$$

The **efficiency of the Otto cycle** (gasoline engine) depends on the compression ratio (V_1/V_2) and the ratio of molar specific heats(γ).

$$e = 1 - \frac{1}{(V_1/V_2)^{\gamma-1}} \quad \text{(Otto cycle)} \tag{22.7}$$

Entropy, S, is a thermodynamic variable which characterizes the degree of disorder in a system.

The **change in entropy** of a system depends on the process which takes the system between the initial and final states. When energy is absorbed by the system, dQ_r is positive and the entropy of the system increases; when energy is expelled by the system, dQ_r is negative and the entropy of the system decreases.

When a system moves along a reversible path between infinitesimally separated equilibrium states, it is assumed that T remains constant.

Reversible process

$$dS = \frac{dQ_r}{T} \tag{22.8}$$

When a system moves between initial and final states via a finite process, T may not be constant. However, the change in entropy is the *same for all paths* connecting the two states.

Finite process

$$\Delta S = \int_i^f dS = \int_i^f \frac{dQ_r}{T} \tag{22.9}$$

When a system is carried around a reversible cycle the change in entropy is identically zero. *The integration must be carried out around a closed path.*

Reversible cycle

$$\oint \frac{dQ}{T} = 0 \tag{22.10}$$

The adiabatic free expansion of a gas is an example of an irreversible process. The free expansion results in an increase in the entropy (and disorder) of the gas.

Irreversible process
(Example: Free expansion of a gas)

$$\Delta S = nR\ln\left(\frac{V_f}{V_i}\right) \tag{22.11}$$

The microscopic definition of entropy is based on the number of *microstates* corresponding to a given *macrostate.* A greater number of microstates corresponds to increased level of entropy of the macrostate.

Entropy of a macrostate

$$S \equiv k_B \ln W \tag{22.14}$$

W = number of microstates